

# CATS<sup>+</sup>

## Competishun Advance

### Training Sessions

(Lecture - )



# Column Matching

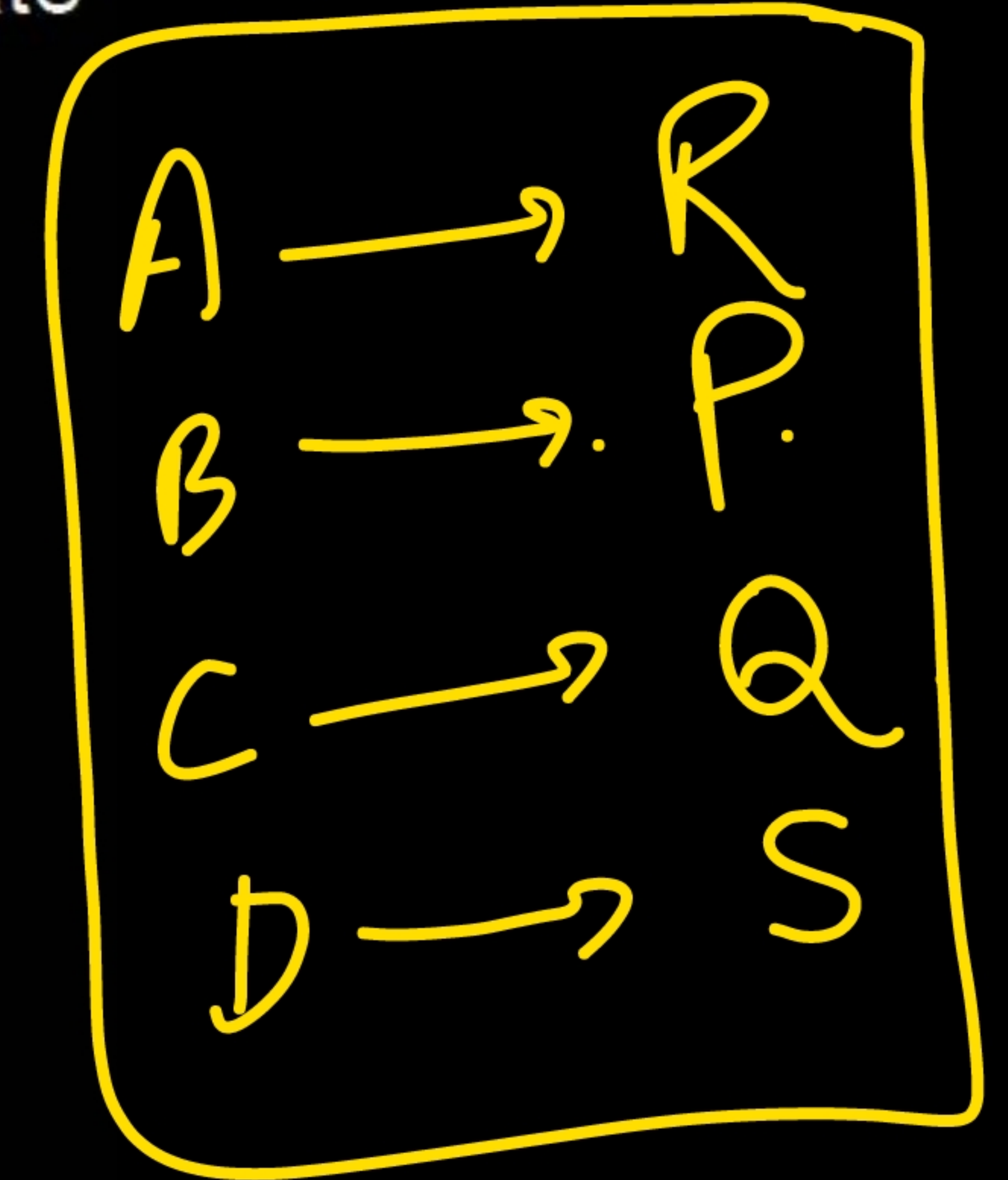
A variable plane cuts x-axis, y-axis and z-axis at points A, B and C respectively such that volume of tetrahedron OABC remains constant and equal to 32 cubic units where O is origin of the coordinate system.

## Column-I

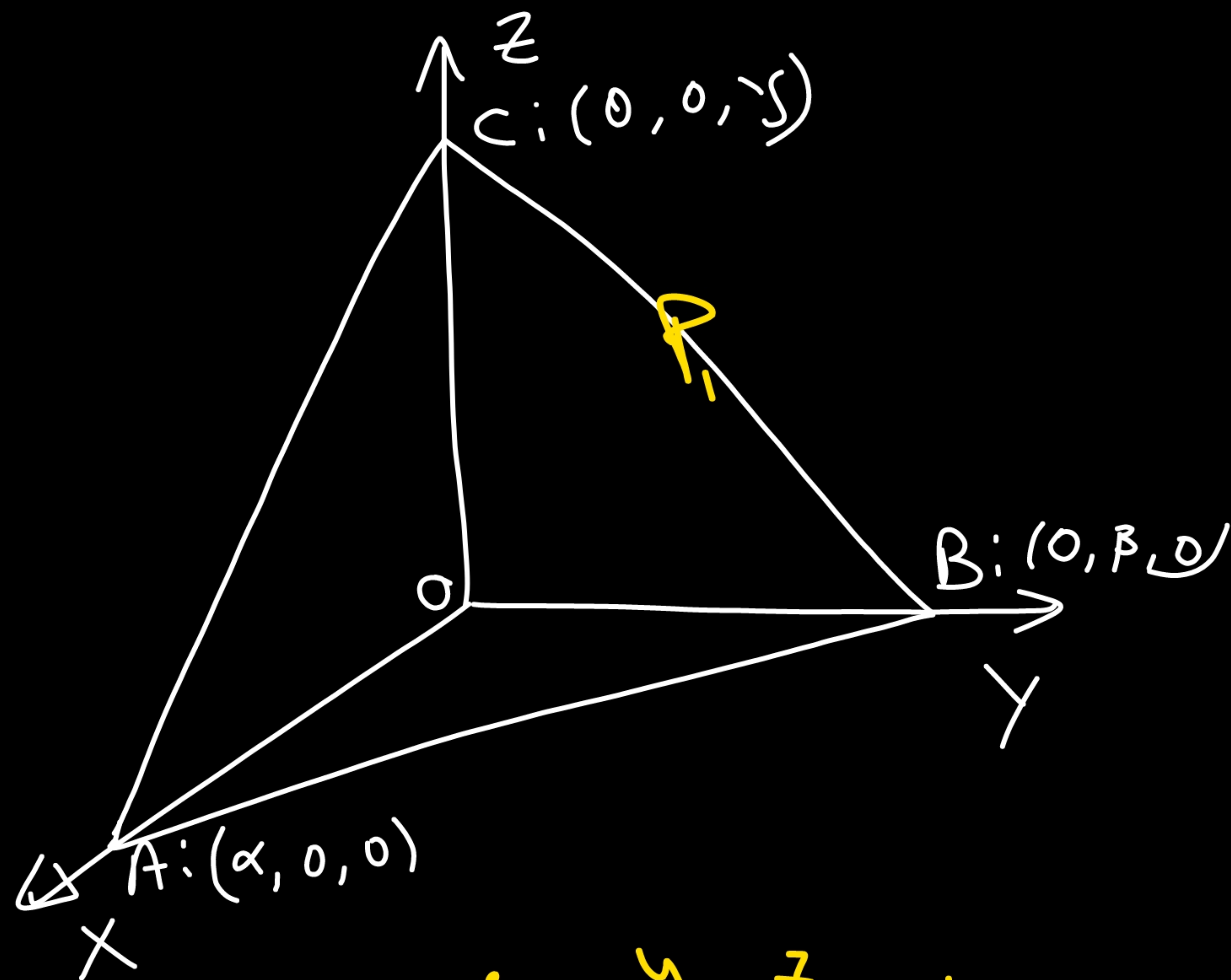
- (A) Locus of centroid of tetrahedron is
- (B) Locus of point equidistant from O, A, B, C is
- (C) Locus of foot of perpendicular from origin to plane is
- (D) If PA, PB and PC are mutually perpendicular then locus of P is

## Column-II

- (p)  $xyz = 24$
- (q)  $(x^2 + y^2 + z^2)^3 = 192 xyz$
- (r)  $xyz = 3$
- (s)  $(x^2 + y^2 + z^2)^3 = \underline{1536 xyz}$







$$P_1 = \frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$$

$$\text{Volume of } OABC = 32 = \frac{1}{6} |\alpha \beta \gamma|$$

$$\Rightarrow \underline{\alpha \beta \gamma = \pm 32 \times 6.}$$

$$G_1(x_1, y_1, z_1)$$

$$\begin{aligned} \bar{G} &= \frac{\sum \bar{a}}{4} = \frac{0 + \alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}}{4} \\ &= \left(\frac{\alpha}{4}\right) \hat{i} + \left(\frac{\beta}{4}\right) \hat{j} + \left(\frac{\gamma}{4}\right) \hat{k} \end{aligned}$$

$$x_1 = \frac{\alpha}{4} \Rightarrow \alpha = 4x_1$$

$$\beta = 4y_1$$

$$\gamma = 4z_1$$

$$\begin{aligned} \cancel{4x_1} \cdot \cancel{4y_1} \cdot \cancel{4z_1} &= \pm 32 \cdot 6^3 \\ x_1 y_1 z_1 &= \pm 3 \rightarrow \boxed{xyz = \pm 3} \end{aligned}$$



② Let required pt is  $P(x, y, z)$

$$OP = PA = PB = PC$$

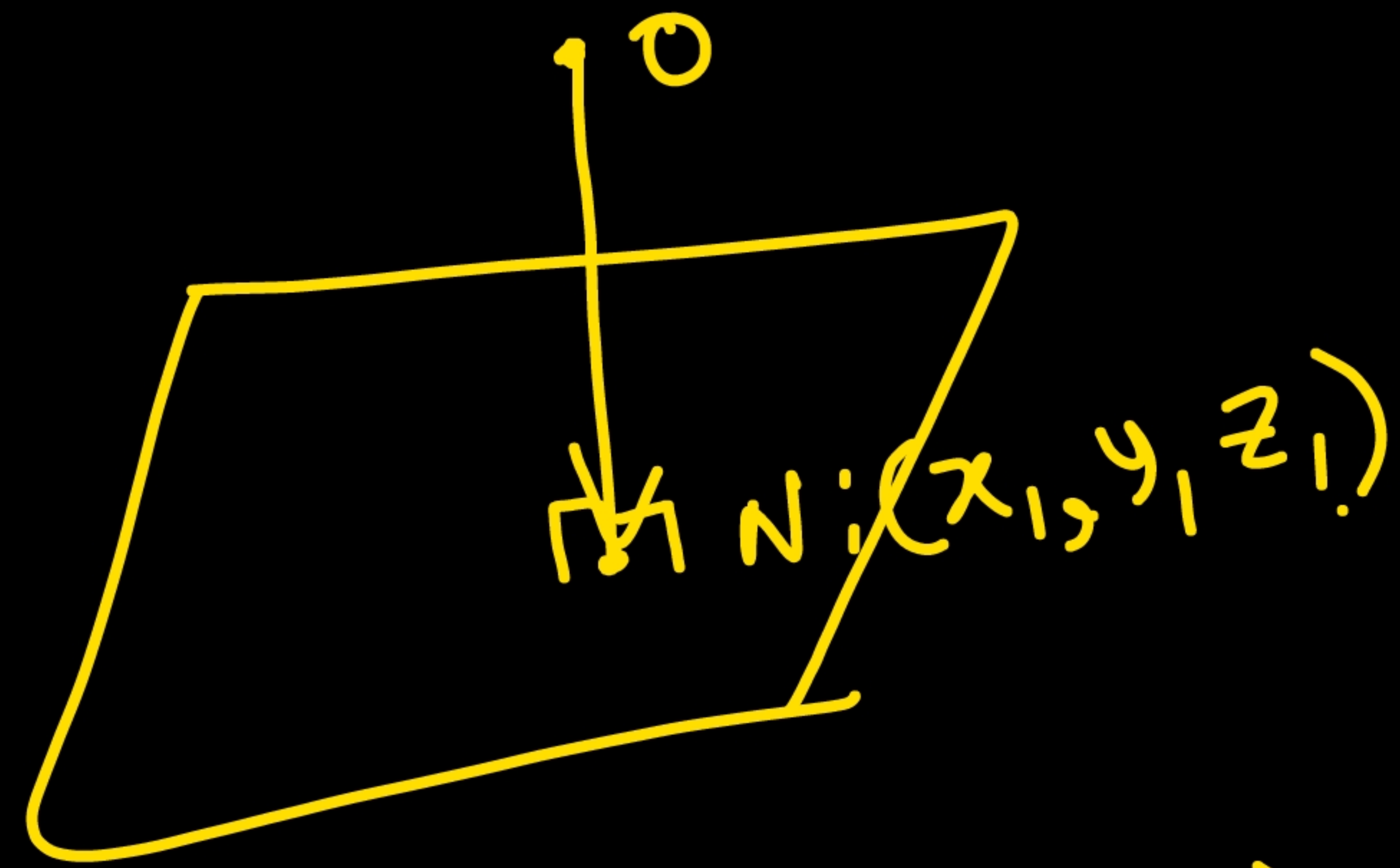
$$OP^2 = PA^2$$

$$\cancel{(x-0)^2 + (y-0)^2 + (z-0)^2} = \cancel{(x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2}$$

$$\begin{aligned} x^2 - 2\alpha x_1 &\Rightarrow \alpha = 2x_1 \\ \parallel^y \quad \beta &= 2y_1 \\ \parallel^z \quad \gamma &= 2z_1 \end{aligned}$$

$$\begin{aligned} \alpha\beta\gamma &= \pm 32x_1y_1z_1 \\ &= 2x_1 \cdot 2y_1 \cdot 2z_1 \Rightarrow \end{aligned}$$

$$xyz = \pm 24$$



$$x_1(x-x_1) + y_1(y-y_1) + z_1(z-z_1) = 0$$

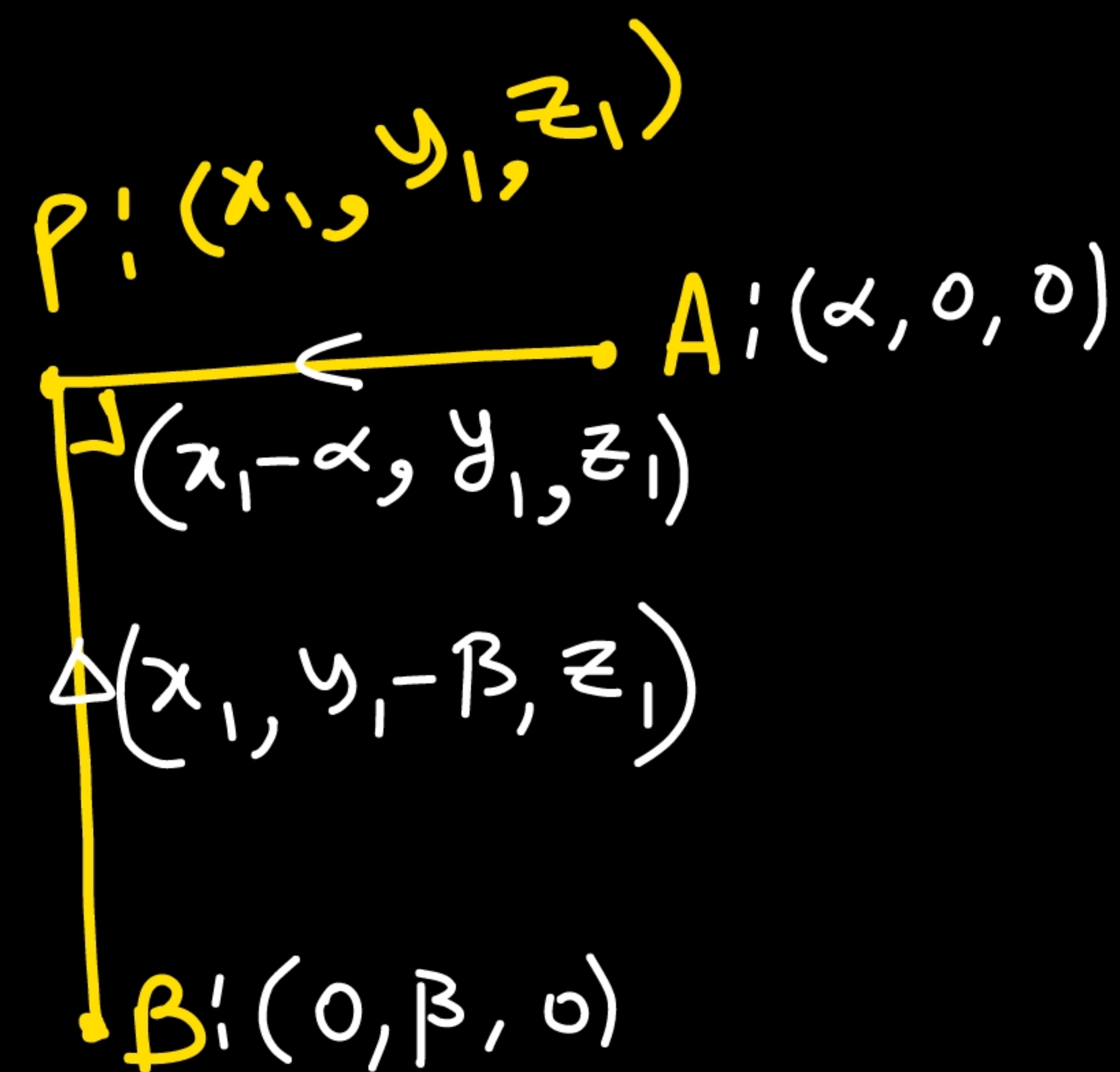
for A  $\rightarrow x_1x - x_1^2 - y_1^2 - z_1^2 = 0$   
 Put  $y=z=0$   $x = \frac{\sum x_1^2}{x_1} = \alpha$   
 $\parallel^y \quad \beta = \frac{\sum x_1^2}{y_1} \quad \parallel^z \quad \gamma = \frac{\sum x_1^2}{z_1}$

$$\alpha\beta\gamma = \pm 32 \cdot 6 \Rightarrow \frac{(\sum x_1^2)^3}{x_1 y_1 z_1} = \pm 192$$

$$\boxed{(x^2 + y^2 + z^2)^3 = 192xyz}$$



①



$$\overrightarrow{PA} \cdot \overrightarrow{PB} = 0$$

$$x_1(x_1 - \alpha) + y_1(y_1 - \beta) + z_1(z_1) = 0$$

$$x_1^2 + y_1^2 + z_1^2 = \alpha x_1 + \beta y_1$$

$$\begin{aligned} \Rightarrow \alpha x_1 + \beta y_1 &= \sum x_1^2 \\ \beta y_1 + \gamma z_1 &= \sum x_1^2 \\ \gamma z_1 + \alpha x_1 &= \sum x_1^2 \end{aligned}$$

$$\alpha x_1 = \beta y_1 = \gamma z_1 = \left( \frac{\sum x_1^2}{2} \right)$$

$$\alpha_1 = \frac{\sum x_1^2}{2x_1}, \quad \beta_1 = \frac{\sum x_1^2}{2y_1}$$

$$\gamma = \frac{\sum x_1^2}{2z_1}$$

Given  $\alpha\beta\gamma = \pm 192$

$$\frac{(\sum x_1^2)^3}{8x_1 y_1 z_1} = \pm 192$$

$$\boxed{(\sum x^2)^2 = \pm 1536xyz}$$

$$\begin{aligned} x + y &= \lambda \\ y + z &= \lambda \\ z + x &= \lambda \\ x = y = z &= \frac{\lambda}{2} \end{aligned}$$



## Paragraph

Let  $z$  and  $w$  are two complex numbers such that  $|z| \leq 1$ ,  $|w| \leq 1$  and  $|z + iw| = |z - i\bar{w}| = 2$

where  $i = \sqrt{-1}$

1. Which of the following is true ?

(A)  $|z| = |w| = \frac{1}{2}$

(B)  $|z| = \frac{1}{2}$ ,  $|w| = \frac{3}{4}$

(C)  $|z| = |w| = \frac{3}{4}$

☒ (D)  $|z| = |w| = 1$

2. Which of the following is true ?

(A)  $\operatorname{Re}(z) = \operatorname{Re}(w)$

(B)  $\operatorname{Im}(z) = \operatorname{Im}(w)$

(C)  $\operatorname{Re}(z) = \operatorname{Im}(w)$

☒ (D)  $\operatorname{Re}(w) = \operatorname{Im}(z)$

3. Number of pairs  $(z, w)$  satisfying above conditions is/are

(A) 1

☒ (B) 2

(C) 4

(D) indeterminate



$$|z| \leq 1$$

$$|\omega| \leq 1 \Rightarrow |i\omega| \leq 1 \Rightarrow |k| \leq 1$$

$$|z + i\omega| = 2 \Rightarrow |z + k| = 2$$

$$\begin{cases} |z - i\bar{\omega}| = 2 \\ = |\bar{z} + i\omega| \end{cases} \Rightarrow |\bar{z} + k| = 2$$

$$i\omega = k$$

$$k = \begin{cases} z \\ \bar{z} \end{cases} \Rightarrow |z + \bar{z}| = 2$$

$$|2\operatorname{Re}(z)| = 2$$

$$|\operatorname{Re}(z)| = 1$$

$$\operatorname{Re}(z) = \pm 1$$

$$|\omega| = |\bar{\omega}|$$

$$2 = |z + k| \leq |z| + |k| \leq 1 + 1 = 2$$

$\Rightarrow$  At both places equality must hold.

$$|z + k| = |z| + |k|$$

$$\hookrightarrow \operatorname{Arg}(z) = \operatorname{Arg}(k)$$

$$|z| = 1$$

$$\textcircled{2} |k| = 1$$

$$z = k$$

$$z = i\omega$$

$$\Rightarrow \operatorname{Im}(z) = 0$$

$$\Rightarrow z = 1 \textcircled{1} \omega = -i$$

$$\textcircled{2} z = -1 \textcircled{2} \omega = i$$

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

$\downarrow$   
for equality  
 $z_1$  &  $z_2$  must  
have same  
argument

$z_1$   $z_2$



Ques If the curves  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  and  $2xy = c^2$  have a common tangent then

(A)  $c^2 = |a| |b|$

(B)  $c^2 \leq |a| |b|$

✓ (C)  $c^2 \geq |a| |b|$

(D)  $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$

Soln

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

↓

C.T. →  $y = mx \pm \sqrt{a^2 m^2 + b^2}$

↓

$$2xy = c^2$$

$$2x(mx \pm \sqrt{a^2 m^2 + b^2}) = c^2$$

$$\Rightarrow D = 0.$$

$$2mx^2 \pm 2\sqrt{a^2 m^2 + b^2}x - c^2 = 0$$

$$\Rightarrow D = 0.$$

$$4(a^2 m^2 + b^2) = 4 \cdot 2m \cdot (-c^2)$$

$$a^2 m^2 + 2c^2 m + b^2 = 0.$$

$$D \geq 0.$$

$$4c^4 - 4 \cdot a^2 b^2 \geq 0$$

$$\boxed{c^2 \geq |ab|}$$



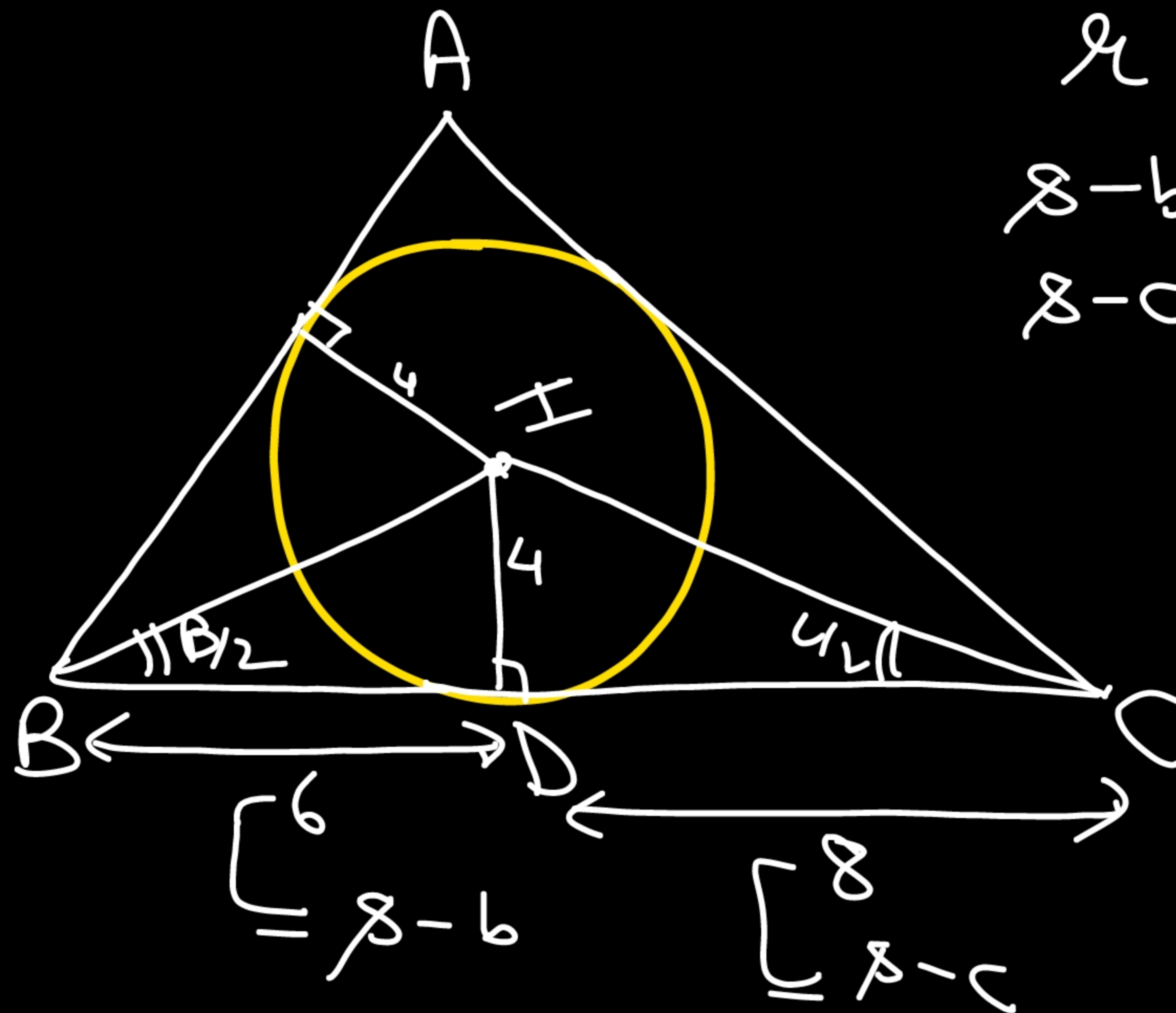


(A) trian

~~(C) perin~~

~~(B) tan~~

~~(D)~~ are



$$\mu = 4$$

$$8 - 2 = 6.$$

$$8 - c = 8$$

$$\tan\left(\frac{\beta}{2}\right) = \frac{4}{6} = \frac{2}{3} = \frac{4}{6}$$

$$\tan\left(\frac{c}{2}\right) = \frac{4}{8} = \frac{1}{2}$$

$$\sum_{i=1}^n \tan\left(\frac{A_i}{2}\right) \tan\left(\frac{B_i}{2}\right) = 1$$

$$\lambda \cdot \frac{2}{3} + \lambda \cdot \frac{1}{2} + \frac{1}{3} = 1 \Rightarrow \lambda \left( \frac{7}{6} \right) = \frac{2}{3}$$

$$\lambda = \frac{4}{7}$$

$$- \left( \tan \left( \frac{A}{2} \right) \right) = \frac{4}{7}$$

A B C D



$$\tan(B/2) = 2/3 < 1$$

$$\Downarrow$$

$$\frac{B}{2} < \frac{\pi}{4}$$

$$B < \frac{\pi}{2}$$

$$h = (s-a) \tan(A/2)$$

$$s-a = \frac{4}{(4/7)} = 7$$

$$s-a = 7$$

$$s-b = 6$$

$$s-c = 8$$

$$s = 21 \checkmark$$

$$a+b+c = 2s = 42$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{\frac{21}{\cancel{7}} \cdot \frac{7 \cdot 6}{\cancel{12}} \cdot 8}$$

$$= 42 \times 2 = \underline{\underline{84}} \quad \underline{\underline{\text{Unit}^2}}$$



Ques

$$\sum_{k=1}^{10} \frac{(-1)^{k-1} {}^{10}C_k}{k} =$$

(A)  $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{11}$

(C)  $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{9}$

(B)  $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{10}$

(D) None of these

~~Sol~~

$$\sum_{k=1}^{10} (-1)^{k-1} \cdot \frac{{}^{10}C_k}{k}$$
$$= \frac{{}^{10}C_1}{1} - \frac{{}^{10}C_2}{2} + \frac{{}^{10}C_3}{3} - \dots$$

$$(1-x)^{10} = {}^{10}C_0 - {}^{10}C_1 x + {}^{10}C_2 x^2 - \dots$$

$$\begin{aligned} & \frac{1}{x} \left[ {}^{10}C_1 x - {}^{10}C_2 x^2 + {}^{10}C_3 x^3 - \dots \right] = 1 - (1-x)^{10} \\ & \left[ {}^{10}C_1 - {}^{10}C_2 x + {}^{10}C_3 x^2 - \dots \right] = \frac{1 - (1-x)^{10}}{x} \end{aligned}$$

$$\begin{aligned} \int_0^1 R. Sum &= \int_0^1 \frac{1 - (1-x)^{10}}{x} dx = \int_0^1 \frac{1 - x^{10}}{1-x} dx \\ &= \int_0^1 (1 + x + x^2 + \dots + x^9) dx = \left[ 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{10} \right] \end{aligned}$$



Que

The value of the expression  ${}^nC_0 4^n C_n - {}^nC_1 4^{n-3} C_n +$

${}^nC_2 4^{n-6} C_n - {}^nC_3 4^{n-9} C_n + \dots$  to  $(n+1)$  terms is

(A) 1

(B)  $4^n$

3. 31

(C)  $3^n$

(D) 0

$$\begin{aligned}
 & {}^nC_0 \cdot 4^n C_n - {}^nC_1 \cdot 4^{n-3} C_n + {}^nC_2 \cdot 4^{n-6} C_n - {}^nC_3 \cdot 4^{n-9} C_n + \dots + (-1)^n \cdot {}^nC_n \cdot 4^{n-3n} C_n \\
 &= \text{Coeff of } x^n \text{ in } \left( {}^nC_0 (1+x)^{4n} - {}^nC_1 (1+x)^{4n-3} + {}^nC_2 (1+x)^{4n-6} - \dots + (-1)^n \cdot {}^nC_n \cdot \underline{(1+x)^n} \right) \\
 &= \text{Coeff } x^n \text{ in } (1+x)^n \left( {}^nC_0 (1+x)^{3n} - {}^nC_1 (1+x)^{3n-3} + {}^nC_2 (1+x)^{3n-6} - \dots \right) \\
 &= \text{Coeff of } x^n \text{ in } (1+x)^n \left( {}^nC_0 \underline{(1+x)^3}^n - {}^nC_1 ((1+x)^3)^{n-1} + \dots \right)
 \end{aligned}$$



$$= \text{coeff of } x^n \text{ in } (1+x)^n \left( \left( \underline{(1+x)^3} - 1 \right)^n \right)$$

$$= \text{coeff of } x^n \text{ in } (1+x)^n (x^3 + 3x^2 + 3x)$$

$$= \text{coeff of } x^n \text{ in } (1+x)^n \cdot \underline{x^n} \cdot (x^2 + 3x + \underline{3})^n$$

$$= \underline{\text{coeff of } x^n} \text{ in } \underbrace{(1+x)^n}_{\substack{\downarrow \\ \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n}} \cdot \underline{x^n} \cdot \left( \underbrace{3^n + a_1x + a_2x^2 + \dots}_{\substack{\uparrow \\ \dots + a_{2n}x^{2n}}} \right)$$

$$= \binom{n}{0} \cdot 3^n = \boxed{3^n} \quad \text{✓}$$



$${}^{25}C_{11} = \text{Coeff of } \underline{x^{11}} \text{ in } (1+x)^{25}$$

$$7 \cdot {}^{35}C_9 = \text{Coeff of } x^9 \text{ in } (1+x)^{35} \cdot 7$$

$$(x-y)^n = {}^nC_0 x^n y^0 - {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 - {}^nC_3 x^{n-3} y^3 + \dots - {}^nC_n x^0 y^n$$

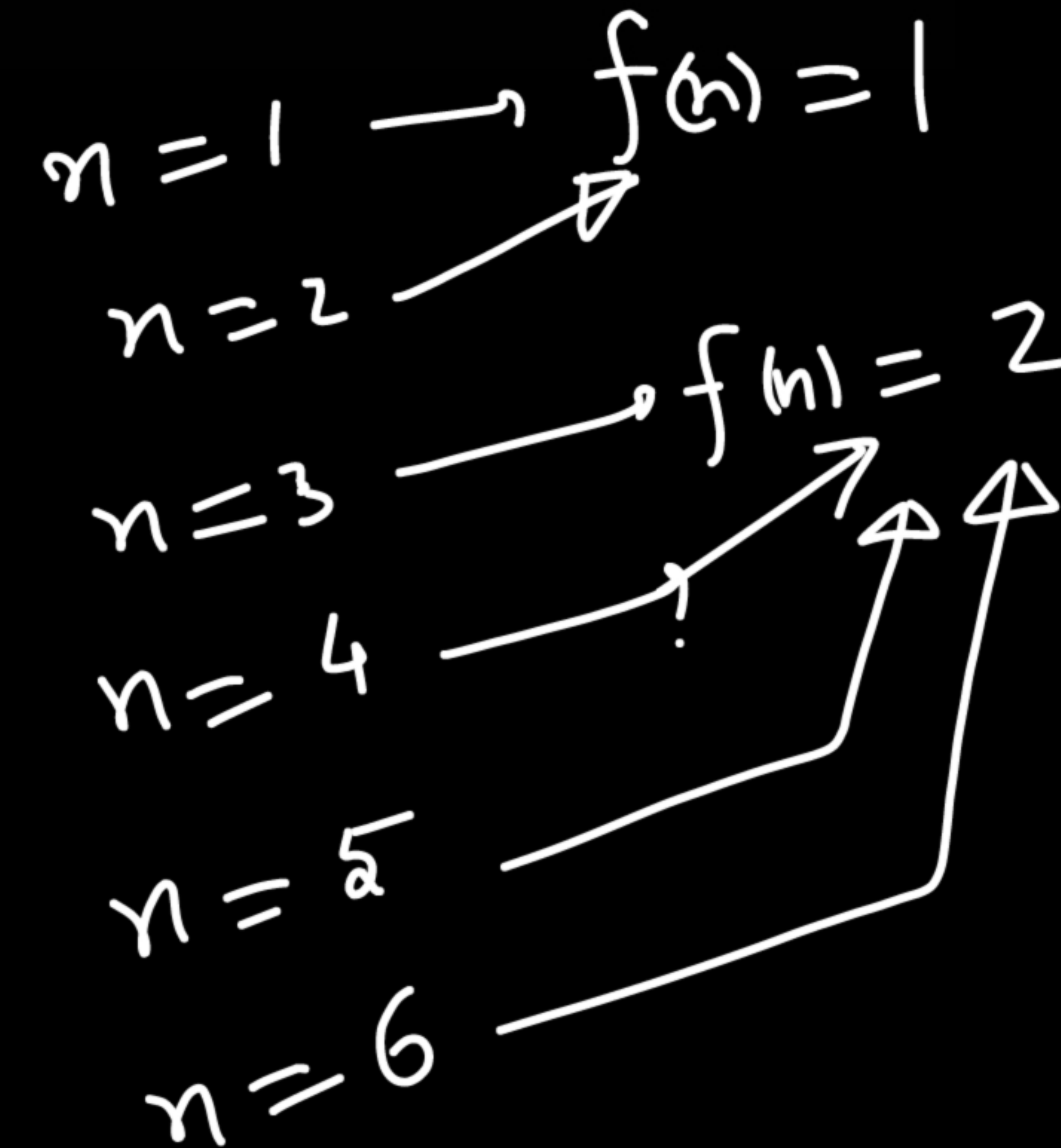


Ques  $f(n) = \left[ \sqrt{n} + \frac{1}{2} \right]$ , where  $n \in \mathbb{N}$  then evaluate  $\sum_{n=1}^{\infty} \frac{2^{f(n)} + 2^{-f(n)}}{2^n}$  (where  $[.]$  denotes greatest integer function)

Sol  $f(n) = \left[ \sqrt{n} + \frac{1}{2} \right], n \in \mathbb{N}.$

$$\sum_{n=1}^{\infty} \frac{2^{f(n)} + 2^{-f(n)}}{2^n}$$

$$\sqrt{5} + 0.5$$



$$\text{Let } \left[ \sqrt{n} + \frac{1}{2} \right] = r$$

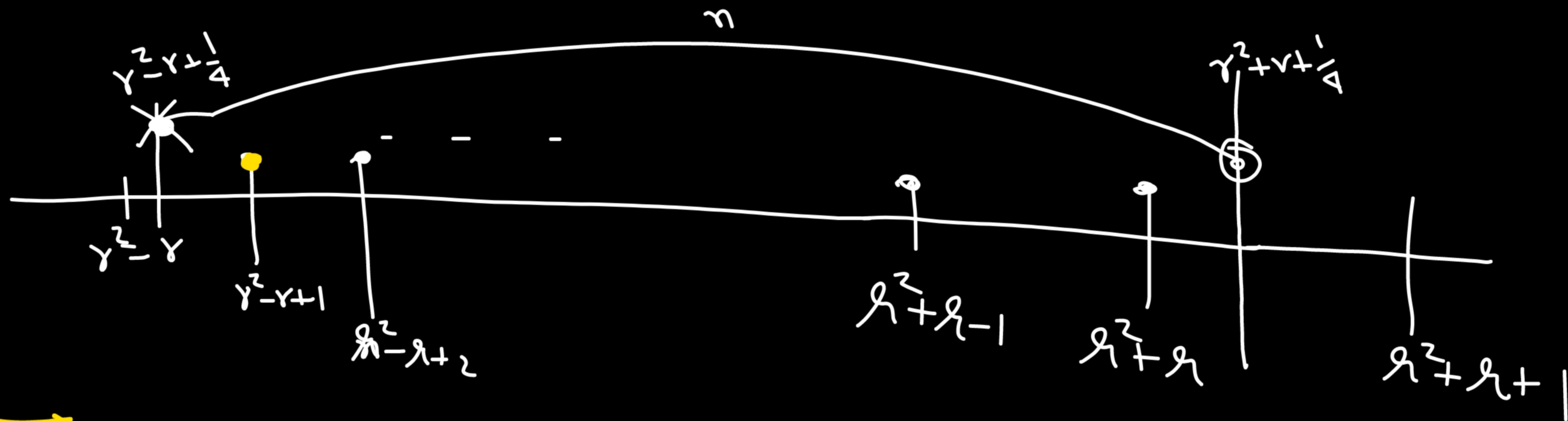
$$r \leq \sqrt{n} + \frac{1}{2} < r+1$$

$$r - \frac{1}{2} \leq \sqrt{n} < r + \frac{1}{2}$$



$$\left(r - \frac{1}{2}\right)^2 \leq n < \left(r + \frac{1}{2}\right)^2$$

$$r^2 - r + \frac{1}{4} \leq n < r^2 + r + \frac{1}{4}$$



$$r^2 - r + 1 \leq n \leq r^2 + r$$

$$f(n) = r \Rightarrow n \in \{r^2 - r + 1, r^2 - r + 2, \dots, r^2 + r\}$$

$$r=1 \rightarrow f(n)=1 \Rightarrow n \in \{1, 2\}$$

$$r=2 \rightarrow f(n)=2 \Rightarrow n \in \{3, 4, 5, 6\}$$



$$\sum_{n=1}^{\infty} \frac{2^{f(n)} + 2^{-f(n)}}{2^n}$$

$$f(n) = r$$

$$n \rightarrow r^2 - r + 1 \rightarrow r^2 + r$$

$$= \sum_{r=1}^{\infty} \left( \sum_{n=r^2-r+1}^{r^2+r} \frac{2^r + 2^{-r}}{2^n} \right) = \sum_{r=1}^{\infty} \left( (2^r + 2^{-r}) \sum_{n=r^2-r+1}^{r^2+r} \left( \frac{1}{2} \right)^n \right)$$

$$= 2r$$

$$= \sum_{r=1}^{\infty} (2^r + 2^{-r}) \left( \left( \frac{1}{2} \right)^{r^2-r+1} \frac{1 - \left( \frac{1}{2} \right)^{2r}}{1 - \frac{1}{2}} \right)$$

$$= \sum_{r=1}^{\infty} (2^r + 2^{-r}) \left( \frac{1}{2} \right)^{r^2-r} \left( 1 - \left( \frac{1}{2} \right)^{2r} \right)$$

$$= \frac{1 - r^n}{1 - r}$$



$$\sum_{r=1}^{\infty} \left( 2^r + 2^{-r} \right) \left( \frac{1}{2} \right)^{r^2-r} \left( 1 - \left( \frac{1}{2} \right)^{2r} \right)$$

$$= 2^r + \frac{1}{2^r}$$

$$= \frac{(2^{2r} + 1)}{2^r} \cdot \frac{1}{2^{r^2-r}} \cdot \frac{2^{2r} - 1}{2^{2r}}$$

$$= \frac{2^{4r} - 1}{2^{r^2+2r}} = 2^{2r-r^2} - 2^{-r^2-2r}$$

$$= 2^{\frac{-r(r-2)}{2}} - 2^{\frac{-r(r+2)}{2}}$$

$$V_r - V_{r+2}$$

$$= \lim_{n \rightarrow \infty} \left( 2^{-1(-1)} + 2^{-0} - 2^{\frac{-n(n+2)}{2}} - 2^{\frac{-(n-1)(n+1)}{2}} \right)$$

$$= 2 + 1 = \boxed{3}$$

~~Q.E.D.~~